SchoolBridge

Technical Design Document v2.0

WhatsApp-First School Operations & Revenue Automation Platform

Version	2.0 — Full Strategic Edition
Date	2025
Status	Internal — Pilot Ready
Audience	Technical team, investors, school directors
Country	Uganda (East Africa expansion planned)

CONFIDENTIAL

1. Executive Summary

SchoolBridge is a WhatsApp-first school operations and revenue automation platform purpose-built for Ugandan schools. It addresses the most painful daily problems school administrators, bursars, teachers, and directors face: delayed fee collection, poor parent communication, manual academic reporting, and fragmented payment reconciliation.

Unlike generic school management software imported from abroad, SchoolBridge is designed from the ground up for the Ugandan school operating environment — integrating with MTN Mobile Money and Airtel Money, supporting UNEB examination report formats, operating reliably on low-bandwidth connections, and communicating with parents through WhatsApp, the platform they already use daily.

> **Strategic Positioning**
> SchoolBridge is not a school management system. It is a school operations, communication, and revenue automation platform. Every feature is designed to either save administrator time, increase fee collection rates, or strengthen the school-parent relationship.

The platform targets Uganda's 20,000+ registered primary and secondary schools, the majority of which still manage fees manually, send paper report cards, and communicate with parents through unreliable individual phone calls.

Metric	Current (Manual)	With SchoolBridge
Fee collection rate	45 – 65%	80 – 95%
Parent communication	Ad hoc phone calls	Automated WhatsApp
Report card generation	3 – 5 days manually	Under 30 minutes
Admin time on fee tracking	15 – 20 hrs/week	Under 3 hrs/week
Payment reconciliation	Manual, error-prone	Automated + audited

2. Problem Statement

Uganda's school sector is large, growing, and chronically underserved by technology. The problems are systemic and well-documented:

2.1 Fee Collection Crisis

The average Ugandan school collects only 50–65% of expected fees per term. The root causes include:

- No automated reminder system — bursars rely on handwritten lists and verbal reminders
- No easy payment channel — parents must travel to school or send cash with children
- No real-time visibility — administrators cannot see who has paid at a glance
- No payment plans — parents with partial ability to pay are treated the same as non-payers

> **Financial Impact**

A school with 500 students at UGX 400,000 per term expects UGX 200,000,000. At 60% collection, it actually receives UGX 120,000,000 — a UGX 80,000,000 shortfall every term. SchoolBridge targets a collection rate above 85%, recovering UGX 50,000,000+ per term for a school of this size.

2.2 Parent Communication Breakdown

- Schools have no reliable broadcast channel — announcements are sent via children or handwritten notes
- Parents have no visibility into academic performance between term report cards
- Fee balance inquiries require a physical visit to the school office
- Emergency alerts cannot reach all parents reliably

2.3 Academic Reporting Inefficiency

- Teachers enter marks in multiple registers and notebooks
- Report cards are handwritten or typed individually — a process taking 3–5 days per class
- No standardised digital record of academic performance over time
- UNEB-format reports (P7, UCE, UACE) require manual reformatting every term

2.4 Administrative Fragmentation

- Student enrolment data lives in physical registers
- Payment records are in separate ledgers from student records
- No single dashboard for school leadership to view operations
- Audit trails are non-existent — fraud and errors go undetected

3. Strategic Positioning & Competitive Moats

SchoolBridge is intentionally positioned not as a generic school management system, but as a revenue automation and operations platform. This framing speaks directly to what school directors and boards care about most: financial sustainability.

3.1 The Four Competitive Moats

Moat	Description	Why competitors cannot copy it
Data Moat	Aggregated anonymised fee collection rates, term performance, and payment behaviour across all schools on the platform	Requires years of real school data; no competitor starts with this
Network Moat	Schools in the same district share term calendars, exam schedules, and UNEB benchmarks. Leaving SchoolBridge means losing access to this network.	Network value compounds with each new school enrolled
Institutional Moat	Official partnerships with diocese networks, school associations, and district education offices. One	Relationship-driven; cannot be replicated with technology alone

	endorsement covers dozens to hundreds of schools.	
Switching Cost Moat	After two terms, a school's full fee history, parent contact database, student academic records, and audit trail live inside SchoolBridge. Migration is extremely disruptive.	Data lock-in is legitimate here because we provide genuine, measurable value

3.2 Uganda-Specific Differentiation

This is SchoolBridge's most powerful and defensible advantage: deep local specificity that foreign SaaS products cannot replicate.

- UNEB format report cards — P7, UCE (S4), and UACE (S6) formats built in. Schools are legally required to issue these formats.
- MTN Mobile Money and Airtel Money — the two dominant payment rails in Uganda, integrated natively
- WhatsApp-first communication — Uganda's most widely used messaging platform, requiring no app installation from parents
- SMS fallback — automated SMS when WhatsApp delivery fails, essential for low-connectivity areas
- Low-bandwidth mode — admin portal functions on 2G/3G connections with offline caching
- Uganda Shilling natively — no currency conversion, no foreign payment processor friction

4. Core System Architecture

SchoolBridge uses a hybrid cloud architecture optimised for reliability, low-latency, and graceful degradation in low-connectivity environments. The system is composed of five major layers:

Layer	Components	Purpose
Presentation	React + TailwindCSS Admin Portal; WhatsApp Business API Interface	User-facing UI for staff; WhatsApp for parents
API Layer	Django REST Framework, JWT Authentication, Role-Based Access Control	Business logic, auth, data validation
Automation Engine	Celery task queue, Redis broker, cron-based scheduler	Automated reminders, reconciliation, reporting
Integration Layer	MTN MoMo API, Airtel Money API, WhatsApp Business API, Africa's Talking SMS	Payments, messaging, fallback SMS
Data Layer	PostgreSQL 16 (primary), Redis 7 (cache/queue), AWS S3 (file storage)	Persistent data, caching, documents

4.1 Multi-Tenancy Model

SchoolBridge is a multi-tenant SaaS platform. Each school is an isolated tenant with its own data namespace, user roles, and configuration. Data isolation is enforced at the database row level using a

school_id foreign key on all school-specific tables, with Django middleware validating tenant context on every request.

- Single shared schema with row-level isolation — cost-efficient and performant
- Each school's data is logically separated and never visible to other schools
- Super-admin accounts can access any tenant for support with a full audit trail

4.2 Offline and Low-Bandwidth Strategy

> **Critical Design Requirement**
> Uganda's internet connectivity is unreliable outside major urban centres. SchoolBridge must function usefully when the internet is slow or unavailable. This is not an optional enhancement — it is a core product requirement.

- Progressive Web App (PWA) with service worker for offline caching of the admin portal
- Critical read operations (student list, fee balances, attendance) available offline using IndexedDB
- Write operations queued locally and synced when connectivity resumes
- WhatsApp delivery failures automatically trigger SMS fallback within 2 minutes
- API responses compressed with gzip; images lazy-loaded and compressed

5. System Modules — Detailed Specification

5.1 Admin Web Portal

The web portal is the primary operational interface for school staff, accessible via any modern browser on desktop, tablet, or smartphone. User roles determine which modules are visible and what actions are permitted.

Role	Permissions
Director	Full access to all modules, analytics, financial reports, staff management, system settings
Bursar	Fee management, payment recording, payment plans, financial reporting, receipts, Mobile Money reconciliation
Teacher	Marks entry for assigned classes/subjects, attendance recording, view own class report cards
Registrar	Student enrolment, profile management, class assignments, parent contact management
Admin	All of the above except financial reports and system settings

Student Management Module

- Student profiles: name, date of birth, gender, class, stream, photo, parent contacts
- Enrolment, transfer, graduation, and withdrawal workflows
- Bulk import via CSV from existing manual registers

- Class and stream management with year-end promotion automation
- Search and filter by name, class, fee status, and academic performance

Attendance Module

- Daily class attendance recorded by teacher on mobile or desktop
- Automated WhatsApp/SMS alert to parent when student is absent without excuse
- Weekly attendance summary reports per class
- Attendance rate tracking with alerts when a student falls below configurable threshold

5.2 Fee Management & Collection Module

This is the highest-value module and the primary driver of adoption. It automates the entire fee lifecycle from invoicing to reconciliation.

Fee Structure Configuration

- Configurable fee items per class level: tuition, lunch, transport, development, examination fees
- Term-based fee schedules (Term 1, 2, 3) with due dates
- Sibling discounts, scholarship deductions, and bursary flags
- Payment plans: define instalment schedules for families with confirmed partial payment arrangements

Payment Recording

- Manual cash payment recording by bursar with auto-generated receipt
- Automatic recording of Mobile Money payments via API webhook — no manual entry
- Each payment linked to a student account with timestamp, amount, channel, and operator reference
- Overpayments automatically flagged and credited to next term

Outstanding Balance Tracking

- Real-time fee balance dashboard: paid, pending, overdue, per student and per class
- Director/bursar sees school-wide collection rate at a glance on login
- Ageing analysis: balances overdue by 0–7 days, 7–30 days, 30+ days
- One-click export to PDF or Excel for board reporting

Automated Reminders

- Fee due date reminder: sent 7 days and 1 day before each payment deadline
- Overdue reminder: sent at 3 days, 7 days, and 14 days after due date
- Custom message templates configurable per school in English, Luganda, or other local languages
- Reminders sent via WhatsApp with SMS fallback; delivery status tracked

Two-Way Parent Interaction

Parents can send WhatsApp messages to the school's SchoolBridge number to self-serve basic requests: BALANCE — returns current fee balance for their child RECEIPT — sends the last payment receipt as a PDF REPORT — sends the most recent term report card HELP — returns a menu of available commands This two-way functionality is a key differentiator from broadcast-only systems and dramatically reduces bursar workload.

5.3 Mobile Money Integration

SchoolBridge integrates with both MTN Mobile Money Uganda and Airtel Money Uganda via their respective APIs to enable parents to pay school fees from their mobile wallets without visiting the school.

Payment Flow

1. Parent receives WhatsApp reminder with unique student payment reference code
2. Parent dials *165# (MTN) or *185# (Airtel), selects Pay Bill, enters SchoolBridge merchant code and student payment reference
3. Mobile money operator processes payment and sends webhook notification to SchoolBridge API
4. SchoolBridge automatically credits the student fee account, generates a receipt, and sends WhatsApp confirmation to parent
5. Payment appears on bursar dashboard in real time — no manual entry required

Feature	MTN MoMo	Airtel Money
API Type	MTN MoMo Open API (v1)	Airtel Africa API
Payment direction	Customer to Business (C2B)	Customer to Business (C2B)
Webhook support	Yes	Yes
Settlement	T+1 bank settlement	T+1 bank settlement
Transaction fee	~1% (operator-side)	~1% (operator-side)

5.4 WhatsApp Communication Layer

The WhatsApp layer is the primary communication channel between the school and parents. It requires a WhatsApp Business API account registered to the school or SchoolBridge on behalf of the school.

Outbound Message Types

Message Type	Trigger	Format
Fee reminder	7 days before due date; 1 day before; 3/7/14 days overdue	Template message with balance and payment instructions
Payment confirmation	Immediate on Mobile Money webhook receipt	Receipt summary with new balance
Absence alert	Same day when student marked absent	Student name, date, school contact
Report card	End of term on director approval	PDF attachment or formatted message
General announcement	Manual broadcast by admin	Free-form text or approved template

Exam timetable	Scheduled by admin	Formatted schedule

WhatsApp Business API Requirements

- A verified WhatsApp Business account registered to the school or SchoolBridge on behalf of the school
- Meta-approved message templates for all automated outbound messages
- 24-hour conversation window for free-form replies after a parent initiates contact
- Webhook endpoint on SchoolBridge backend to receive inbound messages and delivery receipts

5.5 Academic Module

The academic module digitises marks collection, grading, and report card generation with Uganda-specific standards built in.

Marks Entry

- Teachers access their class and subject marks entry form from any device
- Configurable assessment components: mid-term test, end-of-term exam, continuous assessment, practicals
- Marks saved immediately; no submit required — reduces data loss on connectivity drops
- Deputy head teacher or director can view marks entry progress per teacher in real time

Automated Grade Computation

- Totals, weighted averages, and final grades computed automatically on marks entry
- Grading scales configurable: Uganda national scale (D1–F9), percentage, or custom letter grades
- Class ranking and stream ranking computed per subject and overall
- Division computation for S4 and S6 following UNEB aggregation rules

UNEB-Format Report Cards

> **Uganda-Specific Advantage**
> SchoolBridge generates report cards in the exact formats required by the Uganda National Examinations Board for P7, UCE (S4), and UACE (S6) candidates. No foreign school management software offers this. It is the single most requested feature from Ugandan secondary school administrators.

- Primary leaving (P7) report card with all required UNEB fields
- UCE (O-Level) internal mock report in UNEB aggregate format
- UACE (A-Level) internal assessment report with principal and subsidiary pass/fail
- Internal primary/secondary report cards in configurable school-branded format
- Bulk PDF generation — all report cards for a class or entire school in one click
- Digital delivery via WhatsApp to parent + print-ready PDF for school records

5.6 Automation Engine

The automation engine is SchoolBridge's most significant operational differentiator. It runs on a Celery task queue backed by Redis, executing scheduled and event-driven tasks that replace hours of manual administrative work.

Scheduled Automations

Automation	Schedule	Output
Fee due reminders	7 days before, 1 day before fee deadline	WhatsApp/SMS to parents with outstanding balance
Overdue fee escalation	3, 7, 14 days after due date	Escalating reminder messages to parents
Daily attendance alerts	By 10 AM on school days	WhatsApp to parents of absent students
MoMo reconciliation	Daily at 11 PM	Reconciliation report to bursar; flag discrepancies
Weekly collection report	Every Monday morning	PDF summary sent to director
Term report generation	On director approval trigger	Bulk PDF report cards for all students

Event-Driven Automations

- On Mobile Money payment received: credit student account + send receipt to parent
- On student marked absent: trigger parent alert within 30 minutes
- On new student enrolled: send welcome WhatsApp to parent with school contact and payment information
- On fee payment plan approval: schedule instalment reminders for each agreed date

6. Technology Stack

Layer	Technology	Rationale
Frontend	React 18 + TailwindCSS + Vite	Fast development, excellent mobile responsiveness, PWA support
Backend	Python 3.12 + Django 5 + Django REST Framework	Mature ecosystem, strong ORM, excellent for SaaS multi-tenancy
Task Queue	Celery 5 + Redis 7	Reliable async task execution; Redis doubles as cache layer
Database	PostgreSQL 16	ACID compliance, JSON fields for flexible config, proven at scale
File Storage	AWS S3 or DigitalOcean Spaces	Report card PDFs, receipts, student documents

Hosting	AWS / DigitalOcean / Hetzner	Hetzner for cost efficiency; AWS/DO for managed services at scale
WhatsApp	WhatsApp Business API via Africa's Talking or Twilio	Africa's Talking has local Uganda support and UGX billing
SMS Fallback	Africa's Talking SMS	Local Uganda coverage; competitive bulk rates
PDF Generation	WeasyPrint or Playwright (headless)	High-fidelity report card PDF rendering from HTML templates
Authentication	JWT via djangorestframework-simplejwt	Stateless auth suitable for SPA and mobile
CI/CD	GitHub Actions + Docker	Reproducible builds; zero-downtime deployments

7. Security & Compliance

SchoolBridge handles sensitive financial data, student academic records, and parent personal information. Security is a non-negotiable foundational requirement, not an afterthought.

7.1 Application Security

- JWT authentication with short-lived access tokens (15 minutes) and secure refresh tokens (7 days)
- Role-based access control (RBAC) enforced at API middleware level — not just the UI
- Input validation and sanitisation on all user-submitted data using DRF serializers
- CSRF protection on all state-changing requests
- SQL injection prevention through Django ORM parameterised queries (no raw SQL)
- XSS prevention through React's DOM escaping and Content Security Policy headers

7.2 Data Security

- All data encrypted in transit (TLS 1.3) — HTTPS enforced with HSTS
- Database encrypted at rest using cloud provider volume encryption (AES-256)
- Passwords hashed using Argon2
- Mobile Money API credentials stored in environment variables, never in source code
- Automated daily database backups with 30-day retention and point-in-time recovery
- Audit log on all administrative actions: who did what, when, from which IP

7.3 Regulatory Compliance

Compliance Requirements

SchoolBridge operates in compliance with: Uganda Data Protection and Privacy Act 2019 (PDPA) Bank of Uganda guidelines on mobile money payment services Uganda Communications Commission (UCC) regulations for bulk messaging Meta WhatsApp Business Terms of Service and Commerce Policy

- Parent data used only for school communication — never sold or shared with third parties

- Data retention policy: student records retained 7 years post-graduation; deleted on school request after exit

8. Revenue Model

SchoolBridge generates revenue through a combination of subscription, transaction, and service fees. The model is designed to align SchoolBridge's revenue with school success — we earn more when schools collect more.

Revenue Stream	Pricing	When	Per School Avg.
Monthly subscription	UGX 150,000 – 400,000 / month based on student count	Monthly	UGX 250,000
MoMo transaction fee	0.5% of fees processed via platform	Per transaction	UGX 100,000+/term
Setup & onboarding	UGX 200,000 – 500,000 one-time	At sign-up	UGX 300,000
SMS bundle top-up	UGX 50,000 per 1,000 messages	As consumed	UGX 30,000/month
Term data insights report	UGX 100,000 per school per term	End of term	UGX 300,000/year

Revenue Projections

Stage	Schools	Monthly MRR	Annual ARR
Pilot (Month 1–3)	5	UGX 1.25M	UGX 15M
Early Growth (Month 4–9)	25	UGX 6.25M	UGX 75M
Scale (Month 10–18)	100	UGX 25M	UGX 300M
Market Leader (Year 3)	500	UGX 125M	UGX 1.5B (~$400K USD)

Pricing Philosophy

For schools hesitant to commit to a monthly subscription, SchoolBridge offers a fee-success pricing option: no monthly subscription, only 1.5% of fees collected via the platform. This removes the biggest objection and aligns SchoolBridge revenue directly with school collection success. Schools switch to subscription pricing once value is proven.

9. MVP Scope & Pilot Deployment Plan

The MVP focuses on the highest-value, lowest-complexity features that demonstrate the platform's core value proposition to school directors within the first week of use.

9.1 MVP Feature Set (Phase 1 — Months 1–3)

Feature	Priority	Success Metric
Student management and profiles	P0 — Required	All students enrolled in system
Fee structure configuration	P0 — Required	Term fees entered and active
Fee balance dashboard	P0 — Required	Real-time collection rate visible
Manual payment recording + receipt	P0 — Required	All cash payments recorded digitally
WhatsApp fee reminders (outbound)	P0 — Required	First automated reminder sent to 80%+ parents
MTN MoMo payment integration	P1 — High	First MoMo payment auto-reconciled
Airtel Money integration	P1 — High	Both payment rails active
Two-way WhatsApp (BALANCE command)	P1 — High	Parents can self-serve balance queries
Marks entry and report card generation	P2 — Medium	End-of-term reports generated digitally
Attendance and absence alerts	P2 — Medium	Daily alerts sent for absences

9.2 Pilot Success Criteria

The pilot with 3–5 schools will be declared successful when all three of the following metrics are achieved:

- Fee collection rate: equal to or greater than 85% of expected fees for the term
- Parent WhatsApp engagement: equal to or greater than 60% of parents have received and read at least one message
- Administrator time saved: equal to or greater than 5 hours per week reported by bursar or administrator

These three metrics become the core of the SchoolBridge sales pitch to all subsequent schools.

9.3 Pilot Rollout Steps

6. Select 3–5 schools with 200–800 students, mix of primary and secondary, Kampala and peri-urban
7. Two-day onboarding: data migration from existing registers, staff training, WhatsApp Business account setup
8. Week 1–2: manual payment recording only, staff familiarisation with the dashboard
9. Week 3: activate WhatsApp reminders, first automated message broadcast
10. Week 4–6: activate Mobile Money integration, first automated reconciliation
11. End of term: generate first digital report cards, measure all three success metrics
12. Debrief with school directors, collect testimonials, refine based on feedback

10. Product Roadmap

Phase	Timeline	Deliverables
Phase 1	Months 1–3	MVP: student management, fee tracking, manual payments, WhatsApp outbound reminders, basic reporting
Phase 2	Months 4–6	MTN + Airtel MoMo integration, two-way WhatsApp, marks entry, automated reconciliation, attendance module
Phase 3	Months 7–9	UNEB-format report cards (P7/S4/S6), SMS fallback, offline PWA mode, payment plans, analytics dashboard
Phase 4	Months 10–15	School network features (district benchmarks, shared calendars), data insights reports, multi-school director view, React Native mobile app
Phase 5	Months 16–24	Institutional partnerships (diocese, district offices), East Africa expansion (Kenya, Tanzania), third-party integrations API

11. Risk Register & Mitigation

Risk	Level	Mitigation
MTN/Airtel API changes or downtime	**HIGH**	Dual operator integration provides failover. Monitor API status. Maintain relationship with operator partner managers.
WhatsApp Business API policy changes	**MEDIUM**	SMS fallback is always active. Do not depend on WhatsApp for critical payment workflows.
School director resistance to change	**HIGH**	Frame everything around revenue recovery. Lead with fee collection rate improvement, not technology features.
Low-quality internet at schools	**HIGH**	Offline PWA mode, compressed API responses, progressive loading. Tested on 2G connections before launch.
Data privacy concerns from parents	MEDIUM	Clear privacy notice in onboarding. Data only used for school communication. PDPA-compliant.
Competitor entry (local or foreign SaaS)	MEDIUM	Execute on four moats: data, network, institutional, switching cost. Move fast on UNEB integration as primary differentiator.
Key personnel risk (small founding team)	MEDIUM	Thorough documentation. CI/CD automation reduces single-person dependencies. Hire second developer by Month 4.

12. Implementation Team Requirements

The following roles are required to execute the MVP and pilot phases. Roles can be combined in the founding team.

Role	Responsibilities	Start Phase
Full-stack Developer (Lead)	Django backend, React frontend, API integrations, DevOps, CI/CD pipeline	Day 1
Full-stack Developer (2nd)	Mobile Money integration, WhatsApp layer, automation engine, mobile app	Month 3–4
School Relationship Manager	Pilot school acquisition, onboarding, staff training, ongoing support	Month 2
Product / Founder	Product decisions, school sales, investor relations, institutional partnerships	Day 1

Appendix A — Core Data Model

The following are the core database entities and their key fields. Full ERD available on request from the engineering team.

Entity	Key Fields
School	id, name, address, phone, logo, whatsapp_number, momo_merchant_code, created_at
Student	id, school_id, first_name, last_name, dob, gender, class_id, stream, enrolment_date, status
Parent	id, student_id, name, phone, whatsapp_number, relationship, is_primary_contact
FeeStructure	id, school_id, class_level, term, academic_year, items (JSON), due_date, created_at
FeeAccount	id, student_id, term, expected_amount, paid_amount, balance, status, updated_at
Payment	id, fee_account_id, amount, channel (cash/mtn/airtel), reference, operator_ref, recorded_by, timestamp
AcademicRecord	id, student_id, term, subject_id, component_scores (JSON), total, grade, rank, updated_at
Message	id, school_id, recipient_phone, channel, type, content, status, sent_at, delivered_at
AuditLog	id, school_id, user_id, action, entity_type, entity_id, before (JSON), after (JSON), ip, timestamp

Appendix B — API Integration Summary

Integration	Provider	Auth Method	Key Endpoints

MTN Mobile Money	MTN Uganda Open API	OAuth 2.0 + API Key	/collection, /disbursements, /webhook
Airtel Money	Airtel Africa API	OAuth 2.0	/merchant/v1/payments, /webhook
WhatsApp Business	Meta / Africa's Talking	Bearer Token	/messages, /webhooks, /templates
SMS Fallback	Africa's Talking SMS	API Key	/messaging, /delivery-reports

Document Control

Document title	SchoolBridge Technical Design Document v2.0
Owner	SchoolBridge Product Team
Classification	Confidential — Internal and Investor Use Only
Next review	After pilot completion (approximately Month 4)
Contact	schoolbridge.ug	info@schoolbridge.ug

A Note on This Document

This document reflects the full strategic and technical vision for SchoolBridge. Individual sections will be expanded into separate engineering specifications as the team grows. The goal of this document is to give school directors, investors, and technical partners a complete, honest picture of what SchoolBridge is, how it works, and why it will succeed.